

Transposases examined by secondary structure prediction programs.

Family	Element (or protein) analyzed	Active or # copies in genome ¹	From secondary structure, type of DDE/D motif ²	Relevant references ³
IS1	IS1N	>40*	DD(24)E	*Nyman <i>et al.</i> , 1981; Ohta <i>et al.</i> , 2002, 2004;
	ISSto9	5	DD(20)E	Siguier <i>et al.</i> , 2009
IS1595	1. ISPna2	—	DD(36)N	Siguier <i>et al.</i> , 2009
	2. ISH4	—	DD(36)E	Siguier <i>et al.</i> , 2009
	3. IS1016C	—	DD(34)E	Siguier <i>et al.</i> , 2009
	4. IS1595	—	DD(35)N	Siguier <i>et al.</i> , 2009
	5. ISSod11	13	DD(34)H	Siguier <i>et al.</i> , 2009
	6. ISNWi1	—	DD(35)E	Siguier <i>et al.</i> , 2009
	7. ISNha5	—	DD(33)E	Siguier <i>et al.</i> , 2009
	Merlin: MERLIN1_SM	Consensus	DD(36)E	Feschotte, 2004
IS3	IS911	Active	DD(35)E	Polard and Chandler, 1995; Rousseau <i>et al.</i> , 2002
IS481	IS481	~100*	DD(35)E	*Glare <i>et al.</i> , 1990; Chandler and Mahillon, 2002
IS4	IS50R	Active	PDB ID: 1muh DD(β -strand)E	Rezsöházy <i>et al.</i> , 1993; Davies <i>et al.</i> , 2000
IS701	IS701	Active (15*)	DD(β -strand)E	*Mazel <i>et al.</i> , 1991
	ISRso17	7		
ISH3	ISC1359	5	DD(β -strand)E	
	ISC1439A	13		
IS1634	IS1634	Active (~30*)	DD(β -strand)E	*Vilei <i>et al.</i> , 1999
	ISMac5	7		
	ISPlu4	7		
IS5	IS903	Active	DD(65)E	Derbyshire <i>et al.</i> , 1987; Rezsöházy <i>et al.</i> , 1993; Tavakoli <i>et al.</i> , 1997
	PIF/Harbinger: PIFa (<i>Z. mays</i>)	Active	DD(59)E	Zhang <i>et al.</i> , 2001; Kapitonov and Jurka, 2004; Sinzelle <i>et al.</i> , 2008
IS1182	IS660	3	DD(β -strand)E	Takami <i>et al.</i> , 2001
	ISPsy6	14		
IS6	IS6100	Active	DD(34)E	Martin <i>et al.</i> , 1990; Mahillon and Chandler, 1998
IS21	IS21	Active	DD(45)E	Mahillon and Chandler, 1998; Berger and Haas, 2001
IS30	IS30	Active	DD(33)E	Caspers <i>et al.</i> , 1984; Mahillon and Chandler, 1998
IS66	IS679	Active	DD(α -helical?)E	Han <i>et al.</i> , 2001
	ISPsy5	33		
	ISMac8	3		
IS110	IS492	Active	DEDD	Perkins-Balding <i>et al.</i> , 1999;
	IS1111	20	DEDD	Buchner <i>et al.</i> , 2005
IS256	IS256	Active	DD(α -helical)E	Mahillon and Chandler, 1998; Prudhomme <i>et al.</i> , 2002
	MuDr/Foldback (Mutator)	Active	DD(α -helical)E	Eisen <i>et al.</i> , 1994; Babu <i>et al.</i> , 2006; Hua-Van and Capy, 2008
IS630	ISY100	Active	DD(34)E	Doak <i>et al.</i> , 1994; Feng and Colloms, 2007
	Tc1/mariner: Mos1 (<i>D. mauritiana</i>)	Active	PDB ID: 2f7t DD(34)D	Plasterk <i>et al.</i> , 1999; Richardson <i>et al.</i> , 2006
	Zator:	36*	DD(43)E	*Bao <i>et al.</i> , 2009
	Zator-1_HM			
IS982	ISPfu3	5	DD(47)E	Mahillon and Chandler, 1998
IS1380	IS1380A	~100*	DD(β -strand)E	*Takemura <i>et al.</i> , 1991; Chandler and Mahillon, 2002
	piggyBac (<i>T. ni</i>)	Active	DD(β -strand)D	Cary <i>et al.</i> , 1989; Sarkar <i>et al.</i> , 2003; Mitra <i>et al.</i> , 2008
ISAs1	ISAz03	7	DD(β -strand)E/D?	
ISL3	IS31831	Active	DD(α -helical)E	Suzuki <i>et al.</i> , 2006
	IS651	22		
Tn3	Tn3 (<i>E. coli</i>)	Active	DD(α -helical?)E	Grindley, 2002
hAT	Hermes (<i>M. domestica</i>)	Active	PDB ID: 2bw3 DD(α -helical)E insertion	Warren <i>et al.</i> , 1994; Rubin <i>et al.</i> , 2001; Hickman <i>et al.</i> , 2005
CACTA	CACTA1 (<i>A. thaliana</i>) En/Spm ZM	Active	DD(α -helical?)E/D?	Miura <i>et al.</i> , 2001; DeMarco <i>et al.</i> , 2006
P	<i>Drosophila</i>	Active	?	Rio, 2002

<i>Transib</i>	<i>Transib1_AG</i>	Consensus	DD(α -helical)E	Kapitonov and Jurka, 2005; Chen and Li, 2008
	RAG1 (<i>M. musculus</i>)	Active	DD(α -helical)E	Kim <i>et al.</i> , 1999; Landree <i>et al.</i> , 1999; Lu <i>et al.</i> , 2006
<i>Sola</i>	<i>Sola3-3_HM</i>	Multiple copies*	DD(40)E	*Bao <i>et al.</i> , 2009

Adapted from Hickman et al 2010 : Integrating prokaryotes and eukaryotes: DNA transposases in light of structure

¹Information on the number of copies within the host genome was obtained from ISfinder or the reference indicated by the asterisk.

²Where indicated, the secondary structure predicts an insertion domain between β 5 and α 4 with predominantly either β -strands or α -helices.

³Relevant references include reviews or papers that report the results of secondary structure prediction, report sequence alignments or consensus sequences, identify the DDE/D catalytic residues, or demonstrate that the element is active. The association of certain eukaryotic superfamilies to specific IS families is as per Feschotte and Pritham (2005) and references therein.